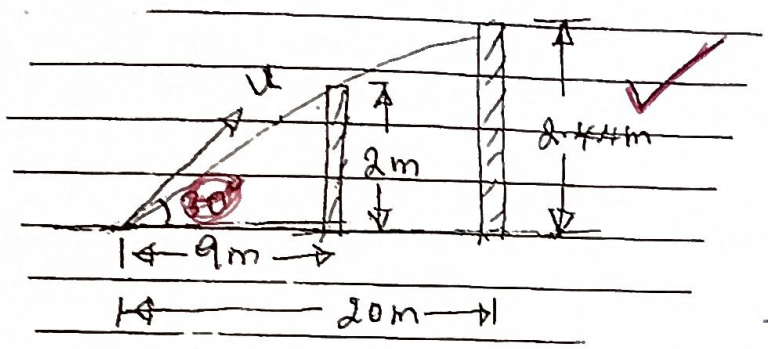
WAKISSHA JOINT MOCK EXAMINATIONS
SCORING GUIDE
Uganda Advanced Certificate of Education
Physics P510/1
July/August 2026

Substitution - 02
θ - 01
U - 01
Units
diagram - 01
Total - 06

Item 1
(a)



$U = ?$
$Q = 30^0$
$X = 9.0m$
$Y = 2.0m$

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

let $\tan\theta = A$, $g = B$

$2 = 9A - 81B$ — (i) ✓

$2.44 = 20A - 400B$ — (ii) ✓

$y = Ax - Bx^2$

$A = 0.3042$

$B = 0.009111$

$\theta = 16.9°$ ✓

$U = 24.26 ms^{-1}$ ✓

05

Using

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

$$2 = 9\tan 30^0 - \frac{(9.81)(9.0)^2}{2u^2\cos^2 30^0}$$

$$2 = 9(0.0577) - \frac{794.6}{1.5u^2}$$

$$u^2 = 165.907$$

$$u = 12.88 m/s$$

Conclusion: ✓

At $u = 12.88 m/s$

When the ball reaches goal post $x = 20\ m$

01

So

$$y = 20\tan 30^0 - \frac{(9.81)(20)^2}{2(165.907)\cos^2 30}$$

$$= 11.547 - 15.767$$

$$= -4.22m$$

This means that the ball landed before it reached the goal posts. The stricker did n't score a goal and Uganda failed to qualify for the round of 16 at PAMOJA AFCON tournament.

03

However if the ball is kicked at angle of 45^0 ;

$$y = 9\tan 45^0 - \frac{(9.81)(9)^2}{2u^2\cos^2 45^0}$$

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- explanation of terminal - 04 - Explain terminal velocity ✓

Radius - 01

Substitution - 01

$F_V = 4.1476N$ - 01

Subst. - $6\pi\eta V_0$ - 01

V_0 - 01

Volume - 01

Total - 10

$U + F_V = mg$ ✓ (04)

$\frac{\rho}{2} \times \frac{4}{3}\pi R^3 g + F_V = mg$ ✓

$1.25 \times \frac{4}{3}\pi \left(\frac{68\times10^{-2}}{2\pi}\right)^3 \times 9.81 + F_V = 0.43 \times 9.81$ ✓

$F_V = 4.1476N$ ✓

$F_V = 6\pi\eta V_0 \rho$

$4.1476 = 6\pi(1.8\times10^{-5}) V_0 \times \left(\frac{68\times10^{-2}}{2\pi}\right)$

$V_0 = 1.13 \times 10^5 ms^{-1}$ ✓

$R = \frac{c}{2\pi} = \frac{68\times10^{-2}}{2\pi}$

$R = 0.1082m$ ✓

$2 = 9 - \frac{794.61}{u^2}$

$u^2 = 118.515$

$u = 10.654 ms^{-1}$

At this speed

$y = 9\tan 45^0 - \frac{9.81(9)^2}{2(113.515)Cos^2 45}$

$= 2m$

Volume = $0.00530977m^3$ from

This means that at 45^0, the stricker would score from the free kick and Uganda would qualify.

03

V - 02 ($50ms^{-2}$)

F_d - 02 (24762.5N)

Recommendation - 03 Any 3

Work done - 02

Total - 09

03

Car motion

Using $V^2 = U^2 + 2@s$

$(30)^2 = 2@(300)$

$@ = \frac{900}{600}$

$@ = 1.5ms^{-1}$

$V = u + at$

$= 0 + (5\times10)$

$= 50ms^{-1}$ ✓

Assuming the car has Q mass of 1000kg

$F = ma$

$= 1000 \times 1.5$

$= 1500N$

distance $s = ut + \frac{1}{2}at^2$

$s = \frac{1}{2}(5)(10)^2$

$= 250m$

$F_d = ma + mg\sin 30$

$= 2500(5) + 2500(9.8)(0.5)$

$F_d = 24762.5N$

Work done $= 24750 \times 250$

$= 6190626J$

05

Work one $W = \frac{1}{2}mv^2 = \frac{1}{2}(1000)(900)$

$= 4.5 \times 10^5 J$

For the drone propeller

Angular speed $w = 2\pi f$

$= 2\pi \times 40$

$= 80\pi\ rad\ s^{-1}$

Centripetal Acceleration

$@ = w^2 r$

$= (0\pi)^2 \times 0.20$

$= 1280\pi^2\ ms^{-2}$

01

(a) Improving the striker's performance.

The sticker should aim at kicking the ball at an angle of 45^0 and ensure sufficient initial speed.

02

For efficient transportation ensure the car engine provides sufficient energy and ensure the tyres grip the road to prevent skidding.

Total = 25

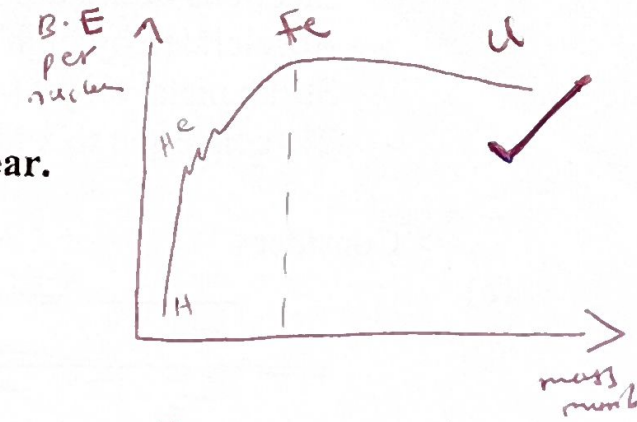
- During a nuclear (reaction) process,

Graph - 01
Explanation - 03
Energy release - 03 → 03
GMT - diagram (01), exp - 01
Background radiation - 01
Activity - 01

em 2

(b) In experiment one;
Binding energy per nuclear is the energy required to remove a nuclear from the nucleus.

i.e BE per nuclear = $\frac{\text{Total binding energy}}{A}$



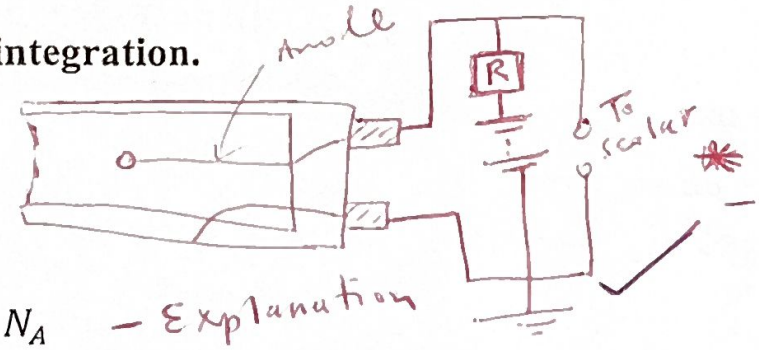
Variation of binding energy permition with that nuclear.
- Increases rapidly for lighter nuclear.
- It peaks around iron (A ≈ 56)
- Decreases slowly for heavy nuclear.

Energy released in fission and fusion.
- Lighter nuclear continue to form heavier nuclear.
- Products have higher binding energy per nuclear (move stable)

- For small nuclei of mass number < 56, the B·E per nucleon ↑ies rapidly and those close to 56 become more stable
- For larger nuclei, B·E per nucleon reduces gradually with ↑se in mass number.
- The stability of nuclei increases

In experiment two.
Geiger nuclear operation;
- Radiations enter the tube through nuclear window.
- Ionize gases (electrons accelerated to a node)
- Avalanche effect produce pulse.
- Pulse counted by the rate meter.

For back ground radiation.
- Counts recorded in absence of source.
- Due to cosmic rays and environmental radiations.



- Explanation

Rate of disintegration.
- $A = \lambda N$
- $\lambda = \frac{In2}{T_{\frac{1}{2}}}$
- $N = \frac{m}{M} \times N_A$

$M = 1g$ $M = 230$ $N_A = 6.02 \times 10^{23}$

$N = \frac{1}{230} \times 6.02 \times 10^{23}$

$\approx 2.62 \times 10^{21}$

$T_{\frac{1}{2}} = 7600 \times 3.65 \times 24 \times 3600 \simeq 2.40 \times 10^{12}$ s

$\lambda = \frac{0.693}{2.40 \times 10^{11}} \simeq 2.89 \times 10^{-13}$ s^{-1}

$A = \lambda N$

$= (2.89 \times 10^{-13})(2.62 \times 10^{21})$

$A = 7.62 \times 10^{8}$ Bq

Energy release

- During nuclear fusion, lighter nuclei combine to form a heavier stable nucleus of higher B·E per nucleon.
- The difference in the mass accounts for energy released.

→ For A > 56, heavy nucleus splits to form lighter nuclei that are more stable.
→ Difference in mass also accounts for energy released.

05

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How x-rays - 03,
photo-electric - 01
Electromagnetic - 02
J.J thompson - 03

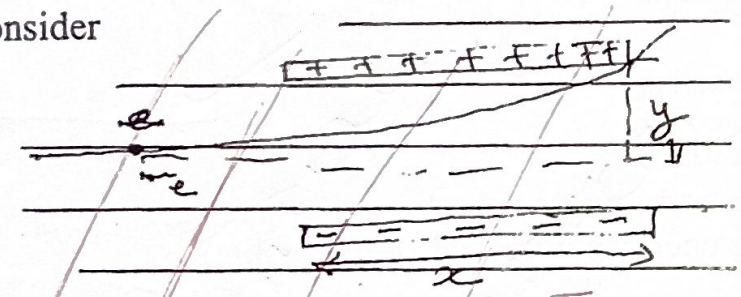
Experiment 3

X – ray production. *(Diagram 02)*

- Elections omitted from the cat bode (thermionic emission)
- Accelerated by high voltage
- Strike metal target (anode)
- 2% converted to x rays and the ret heat energy.

05

(b) Consider



In electric field, electrons experience an electric force and describe a parabolic path.

In a magnetic field, electrons experience a magnetic force and describe a circular path whose direction is perpendicular to the velocity of the electron

Horizontal motion

$S = x$ $\quad a = o\ m5^{-2}$

$u_x = u$

t = 1

From $s = ut + \frac{1}{2}at^2$

$X = ut$

$t = {}^{x}/_{u}$

Vertical nation

$\mathcal{U} = o$ $\quad F_e = ma$

$5^y = y$ $\quad ma = eE$

$A = \frac{eE}{m}$ $\quad a = \frac{eE}{m}$

$y = \frac{1}{2}at^2$

$y = \frac{1}{2}\frac{eE}{m}\left(\frac{x^2}{u^2}\right)$

$y = kx^2$ which in parabolic part

Use of J.J Thomson's experiment in deterring specific change.

- Abeam of electrons is subjected to across field. (Magnetic and electric field)
- At equiriblium

  Magnetic force = Electric force.

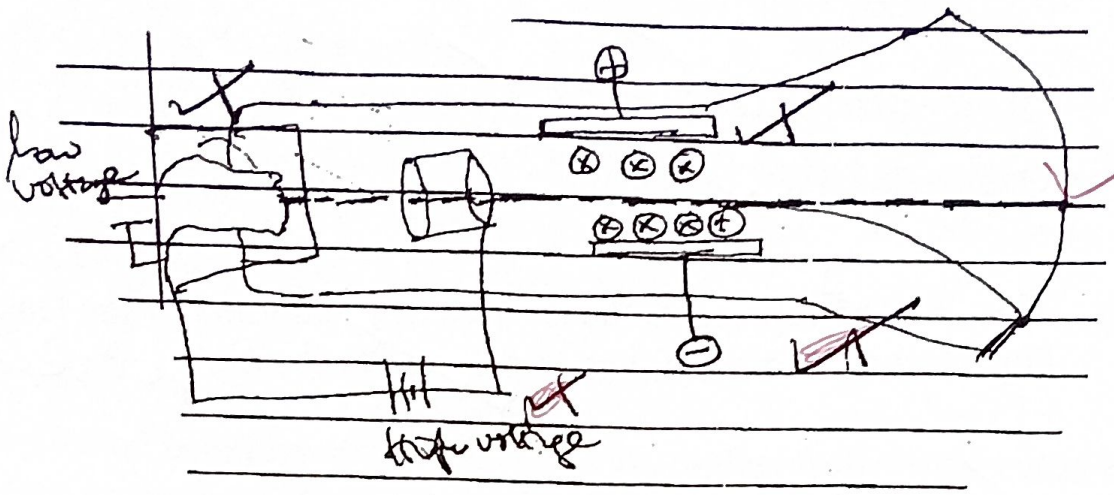
$$Beu = cE \quad \ldots\ldots\ldots\ldots\ldots \text{(i)}$$

$$\mathcal{U} = {}^{E}/_{\beta}$$

$$\text{Also } ev = \tfrac{1}{2}mu^2 \quad \ldots\ldots\ldots\ldots\ldots\text{(ii)}$$

From which

$${}^{e}/_{m} = \tfrac{1}{2}\left[\frac{E^2}{B^2}\right]$$ which gives the specific change of an election.



Accept the five beem experiment.

Application (significance) — CRT, mass spectrometer

05

Part I

Item 3

(a)

Case A

The lens makes formula is

$$\frac{1}{f} = \left(\frac{n\ lens}{nmed} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{fA} = \left(\frac{1.45}{1.80} - 1\right)\left(\frac{1}{50} - \frac{1}{-50}\right)$$

$$\frac{1}{fA} = (1.115 - 1)\left(\frac{2}{50}\right)$$

$$\frac{1}{fA} = \simeq 217.4\ mm$$

$f = 55.6mm$

Case B

Heated lens (nlens = 1.48, n med = 1.30)

$R_1 = 50, R_2 = -50$

$$\frac{1}{fB} = \left(\frac{1.48}{1.30} - 1\right)\left(\frac{1}{50} - \frac{1}{-50}\right)$$

$$\frac{1}{fB} = (1.138 - 1) \times 0.04$$

$$\frac{1}{fB} = 181.2mm.$$

Conclusion: Heating the lens changes the focal length from 217.4mm to 181.2mm.

The neat gained by the oil (Qabsorbed) is given by:

$Qabs = MC\Delta T$

$\Delta T = T_2 - T_1$

$\Delta T = 22.5 - 20.0$

$= 2.5k$

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$Qabs = (0.05) \times (2100) \times 2.5$

$Qabs = 262.5W$

Conclusion: This 262.5W represents the laser energy absorbed by the lens (due to imperfect transmission) and transferred to the oil.

(b) Continuous how calorimeter method:

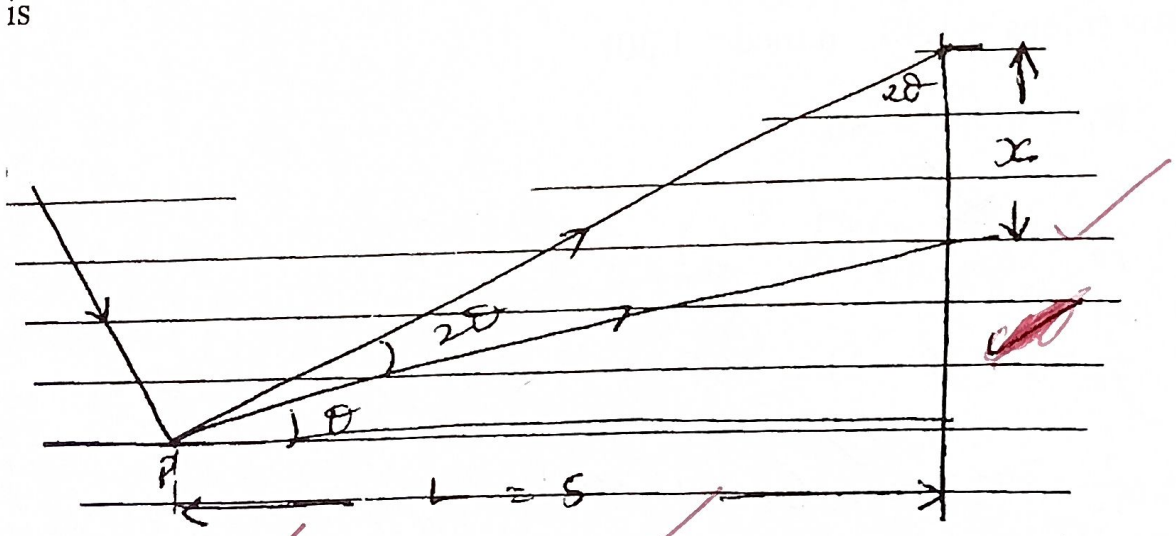
- Steady flow: A liquid (fluid) is passed through a tube or calorimeter vessel at a constant controlled rate.
- In let temperature check (T_1): A thermometer measures the incoming fluid temperature.
- Heat transfer: The liquid or fluid passes through a heater coil, absorbing the heat generated.
- Outlet temperature check (T_2): A second thermometer measures the outgoing warmer fluid.
- Steady state determination: the experimenter waits for the outlet temperature to stop changing, ensuring that all surrounding insulation has heated up and a "steady state" is achieved.
- Data collection and calculation: $C = \dfrac{V_2 I_2 - V_1 I_1}{(m_2 - m_1)(T_2 - T_1)}$

If M – is the mas flow rate.
ΔT – temperature rise.
C – Specific heat capacity.
$Q = MC\Delta T$

Total = 25

Item 4

(a) When the reflecting surface of water is filled through a small angle Q the reflected ray is



$\theta = \frac{x}{2l} = \frac{0.8}{2\times10} = \frac{0.8}{20} = 0.04\ radias$

Wave number $k = \frac{2\pi}{\lambda} = \frac{2\pi}{25} = 0.25$

Amputide $A = \frac{\omega}{k} = \frac{0.045}{0.25} = 0.16m$

The device can generate power if A > 0.155m since A = 0.16m > 0.155m, The project is supported by the lake wave.

$T = 10s$, $f = \frac{1}{10} = 0.1Hz$

The wave equation

$T = 10.0s \quad f = \frac{1}{10} = 0.1Hz$

Using $y = A \sin(2\pi ft - kx)$

$y = 0.16 \sin(2\pi(0.1)t - \frac{2\pi}{25})$

$y = 0.16 \sin(2\pi t - \frac{2\pi}{25})$

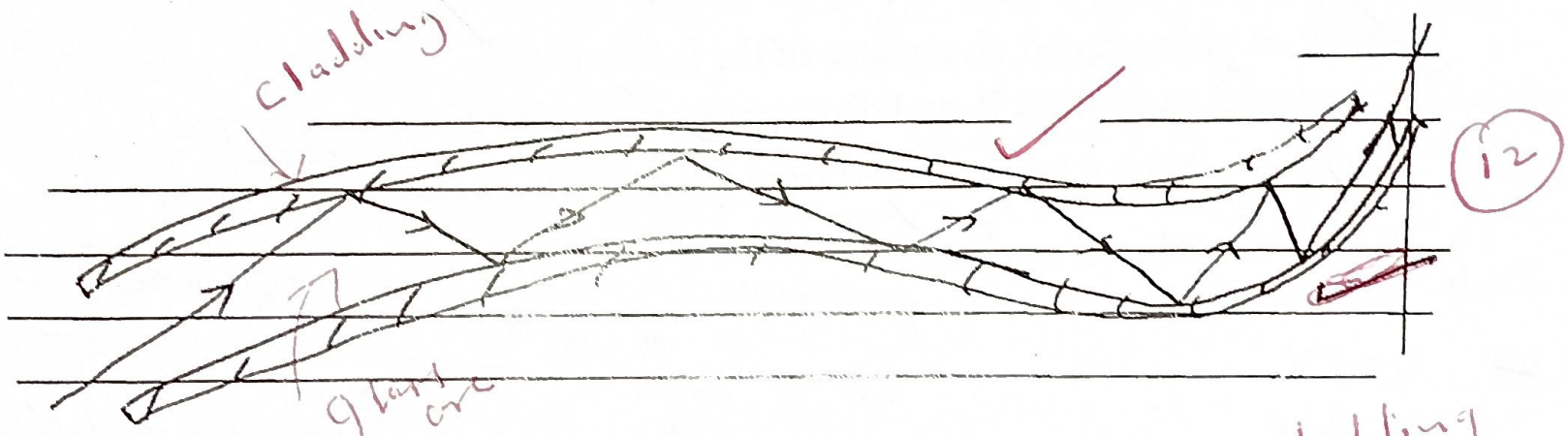
$y = 0.016 \sin[2\pi(0.1)t - 0.25x]$

$y = 0.016 \sin(0.2\pi t - 0.25x)m$

14

(b) The reflected beam from the wave may ~~speed~~ spread. The convex lens the collects the light and converges them. The beam is then focused into the small opening of the optical fiber. Almost all the reflected with is focused in the optical fiber this improves signal strength.

Optical fiber signal transmission



Optical fiber is made of a ~~cave~~ core that has more refractive index than the ~~clotting~~ cladding light is directed at the boundary between the core and the ~~clatting~~ cladding at an angle greater than critical angle.

The light is totally internally reflected multiple total internal reflection without loss of intensity.

This method is accurate because it reduces human energy improves sensitivity.

11

Item 5

(a) Capacitance is the ratio of Magnitude of charge on either mate of the capacitor to the p.d across its plates.

$C = 4\pi\varepsilon_{0r}$

$= 4\pi \times 8.85 \times 10^{-12} \times 7.5 \times 10^{2}$

$= 8.34\ 10^{-2}F \quad 8.34 \times 10^{12} F = 8.34 \mu F$

$Q = CV$

$= 8.34 \times 10^{-12} \times 9 \times 10^{3}$

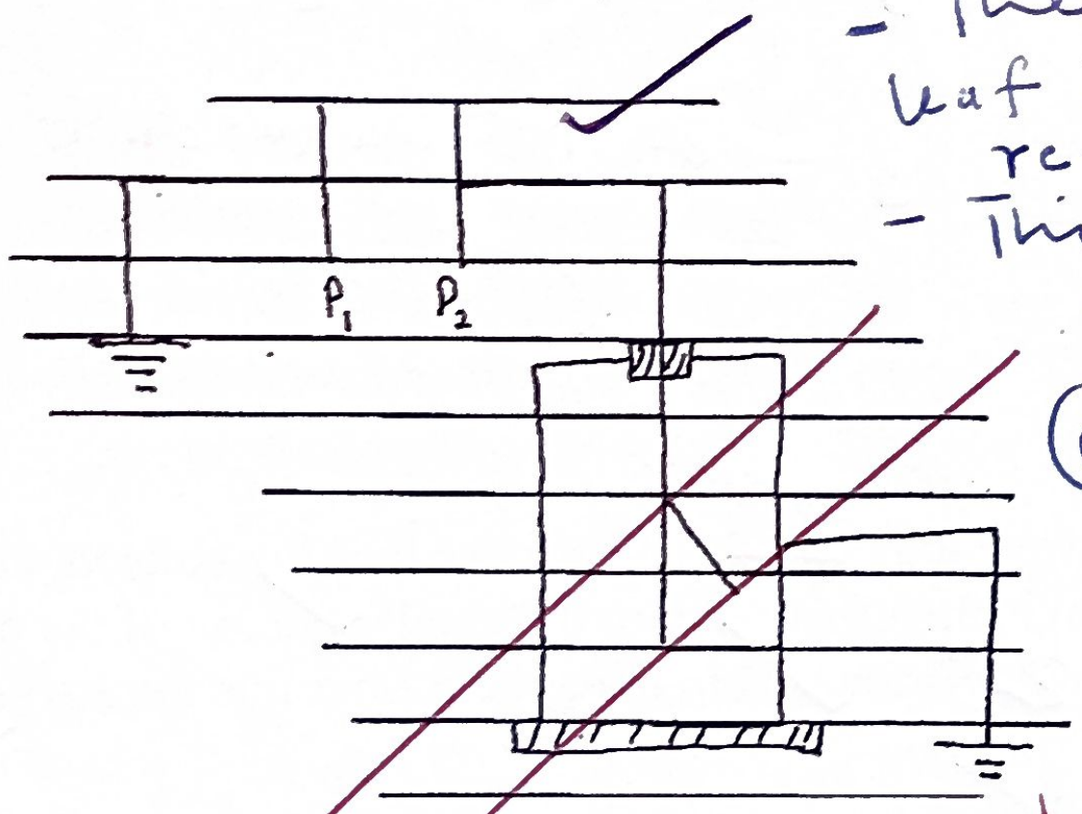
$= 7.5 \times 10^{-8}C \quad 7.51 \times 10^{-11} C$

- keeping area and separation of plates constant, a capacitor is filled with a dielectric material.
- The divergence of the leaf is noted and seen to reduce.
- This means p.d has reduced and capacitance has increased

(03)

$C \propto \frac{1}{V}$, C ↑ as it is dielectric.

(b)



(05)

P_1 and P_2 are metal plates. P_1 is earthed.

Plate P_2 is charged and divergence of the leaf is noted.

P_1 is moved away from P_2 and divergence of the leaf is seen to increase.

This means that p.d has increased from $c = \frac{Q}{V}$ and V increased.

Then C has reduced $c \propto \frac{1}{d}$

For the parallel plate capacitor

$C = \frac{\varepsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 5 \times 10^{-4}}{2 \times 10^{-3}}$

$C = 2.21 \times 10^{-12} F \ (2.21 pF)$

(04)

(a) $Q = CV$

$= \frac{\varepsilon_0 AV}{d}$

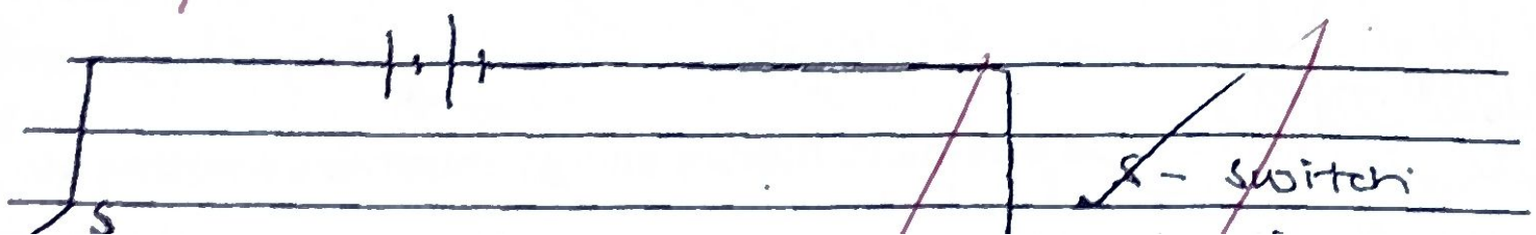
$= \frac{8.85 \times 10^{12} \times 5 \times 10^{-4} \times 6}{2 \times 10^{-3}}$

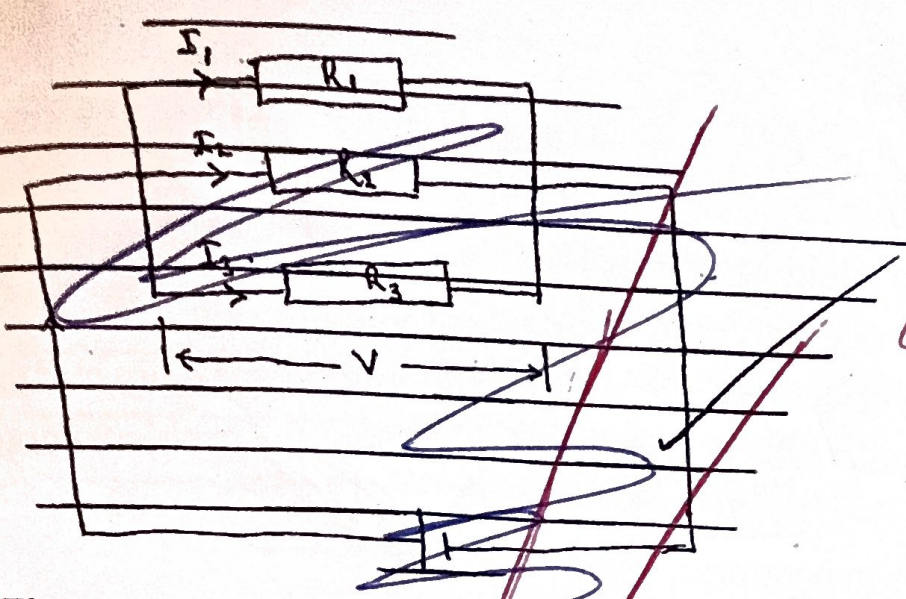
$= 1.33 \times 10^{-11} C.$

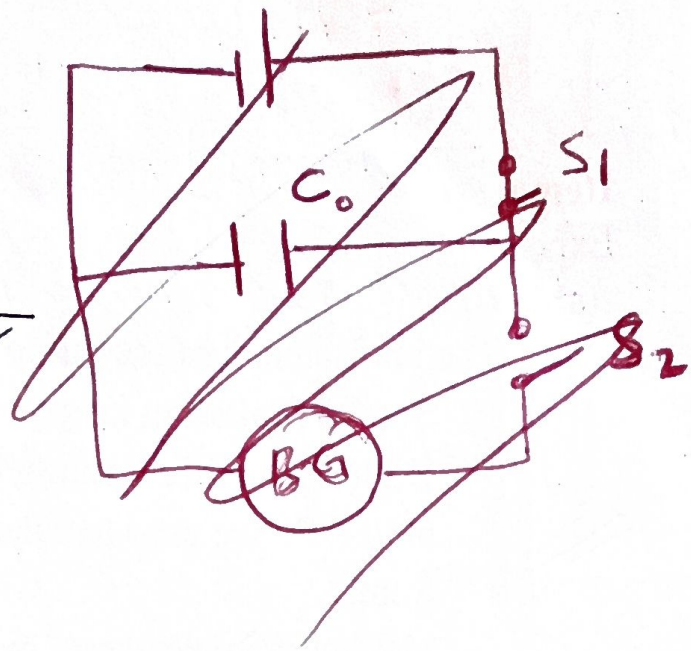
$Q = CV$

$= 2.21 \times 10^{-12} \times 6$

$= 1.33 \times 10^{-11} C$







The current I due to the South divides at the junction A such that different currents I, I and I_2 flow through resistance R_1, R_2 and R_3 respectively.

$$I = I_1 + I_2 + I_3$$

Where $I_1 = \frac{v}{R_1}$, $I_2 = \frac{v}{R_2}$ and $I_3 = \frac{v}{R_3}$.

$$I = \frac{v}{R_1} + \frac{v}{R_2} + \frac{v}{R_3}$$

$$\frac{I}{v} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{I}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_2}$$

R = equivalent resistance.

Power generated = IE But $I = \frac{E}{R+r}$

$$P = \frac{E^2}{R+r}$$

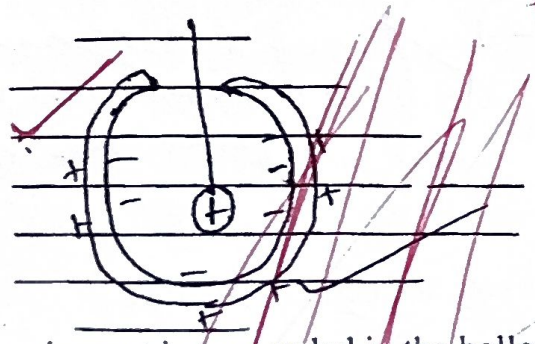
$$= \frac{20^2}{10.0+4.0}$$

P = 28.57W

Efficiency $= \frac{I^2R}{IE} = \frac{R}{R+r} \times 100\%$

$$= \frac{10}{14}$$

$$= 71.4\%$$



The delicate equipment is suspended in the hollow conductor.
Equal and opposite charges are indicated on the inner wall of the hollow condition.
External charges only affects charges on the outside surface.

Total 25

Item 6

Expected Responses

(a) (i)

The inefficiency of the pump could be because of;

- Source limitation i.e generate providing insufficient voltage.
- High impedance limiting effective current flow reducing power delivered to the motor. So we used to calculate the two and find out where the problem comes from.

Maximum induced emf by generator;

$E_0 = NBAw$

$= NBA\ .2\pi f$

$= 500 \times 0.30 \times 0.020 \times 2\pi \times 50$

$E_0 = 942.5V$

$Vms = \frac{E_0}{\sqrt{2}} = \frac{942.5}{\sqrt{2}}$

$= 666.4V$

From our calculation; the generator provides sufficient voltage so the inefficiency could be because of the impedance.

Reatances

Inductive reactances $X_L = 2\pi fL$

$= 2\pi \times 50 \times 0.60$

$= 188.5\Omega$

Capacitive eactance, $X_c = \frac{1}{2\pi(50)(60 \times 10^{-6})}$

$= 53.1\Omega$

Since inductive and capacitive reactance oppose each other in phase.

Net reactance, $X = X_L - X_c$

$= 188.5 - 53.1$

$= 135\ .4\Omega.$

Impedence

$Z = \sqrt{R^2 + x^2}$

$Z = \sqrt{20^2 + x135.4^2}$

$Z = 137\Omega$ (136.9Ω)

The current in the circuit is therefore

$I = \frac{V}{Z} = \frac{666.4}{137} = 4.86x$

If the inductor and capacitor were absent, the current would be

$I = \frac{V}{R} = \frac{666.4}{20} = 33.32x$

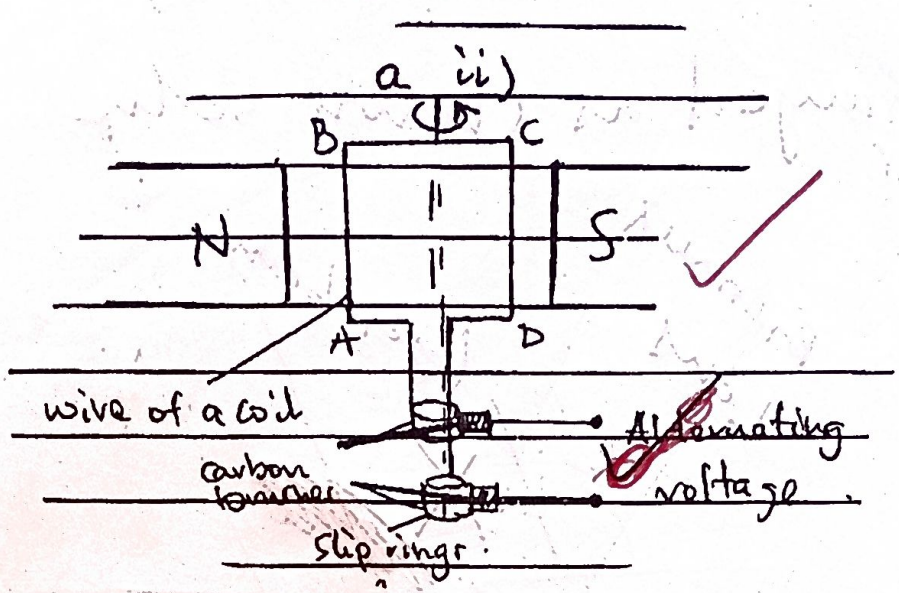
therefore impedance reduces current drawn by the motor from the source leading to inefficiency.

Improvement of efficiency

- Add a capacitor in series to increase capacitive reactance which reduces impedance hence ↑se in current. try 1
- Use of thick copper wires of low resistance to reduce unknown resistance. This reduces power loss due to heating.

17

A.C generator

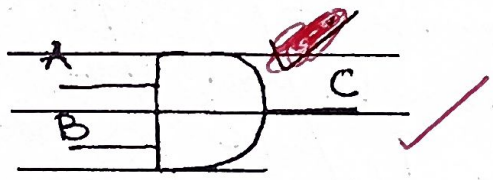


The coil is connected to an external circuit via slip rings and brushes.

The electrical energy in the system is produced by a rotating – coil generator driven by a diesel engine. As the coil rotates in the uniform magnetic field, the magnetic flux linking the coil changes continuously. This changing flux induces an emf in the coil where magnitude depends on the number of turns of the coil, the strength of the magnetic field, area of the coil and angular speed of rotation.

04

(b) The required logic gate is the AND gate which uses two inputs and one output.



Where A represents "water level" input i.e how water level is 1
High water level is 0 and B represented "Voltage level or current level" input
i.e safe is 1
unsafe is 0

04

Truth table

A	B	C
0	0	0
1	0	0
0	1	0
1	1	1

Conclusion

The pump will only be turned on when water level is low and voltage or current levels are safe.

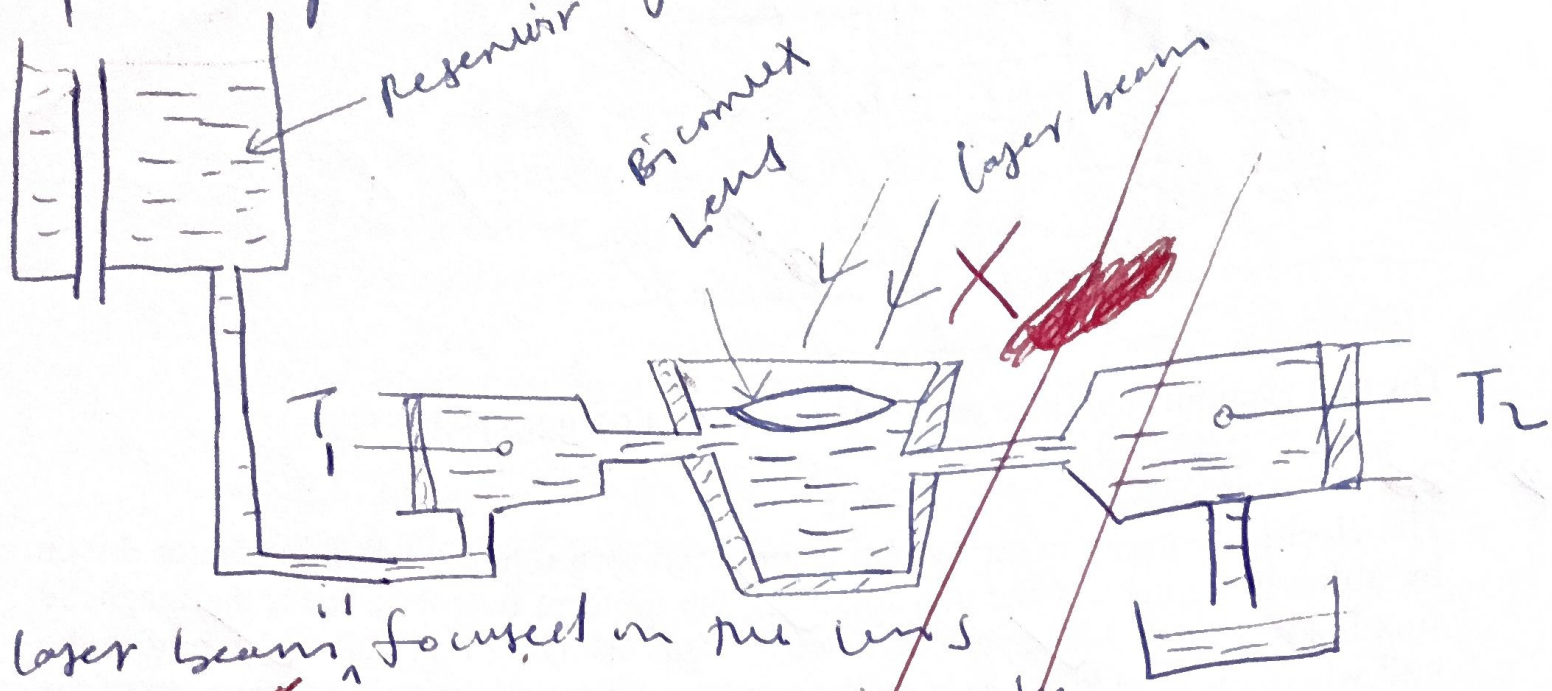
Total = 25

END

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Item 3 (continuous flow method)

- The set up is arranged as shown below.



- A laser beam is focused on the lens continuously such that it continuously absorbs heat.
- The cooling oil with known S.H.C, C_{oil} is passed around the lens at a constant flow rate.
- When steady state is achieved, the inlet temperature T_1 and outlet temperature T_2 are measured and recorded.
- A known mass, m of the cooling oil that passes through the chamber is collected in time t.
- The amount of heat Q absorbed by the lens in every second is determined from the expression

$$Q = \left(\frac{m}{t}\right) C_{oil} (T_2 - T_1)$$

- The heating coil is replaced with the lens having a laser beam continuously incident on it through the glass.
- The heat absorbed by the lens is calculated from;

$$Q = m C_{oil} (T_2 - T_1)$$

where m - mass rate of oil at that instant.
C_{oil} - S.H.C of oil obtained using the known temperatures.
T_1, T_2, inlet and outlet temperatures.